

Linnet

Half-edge graphs and subgraph-first algorithms

1 Overview

Linnet is a graph library for tensor networks, Feynman diagrams, and other workflows where the boundary of a subgraph matters. Its central representation is a half-edge graph. An ordinary internal edge is a pair of half-edges, while an external edge has one dangling half-edge. A node owns incident half-edges, and a subgraph is a selection of half-edges rather than a detached copy of the graph.

Why half-edges

Cutting an internal edge can turn its two halves into boundary edges without changing the degree of either endpoint. This makes operations such as extraction, excision, contraction, sewing, traversal, and cut analysis natural for physics graphs. Linnet algorithms therefore accept subgraph views even when the caller wants to operate on the full graph.

1.1 A working model

A Linnet workflow has four parts:

- construct a graph with `HedgeGraphBuilder`, or parse a DOT graph;
- retain the typed identifiers returned for half-edges, nodes, and edges;
- describe the region of interest with one of the subgraph representations;
- run traversal or mutation operations against that subgraph and the graph's involution.

The involution is the pairing map for half-edges. A paired entry represents an internal edge; an identity entry represents a dangling edge and records its flow. Edge orientation and flow are related concepts, but they are not interchangeable. Consult the API reference for the exact variants and conversion rules in your Linnet release.

Indexes belong to one graph

`Hedge`, `NodeIndex`, and `EdgeIndex` are compact typed indexes, not durable object IDs. Graph edits can change storage and mappings. Keep the result returned by an operation, and do not apply an index or subgraph bitset to an unrelated graph just because its numeric values happen to fit.

1.2 Construction and interchange

The Rust package can be added directly from crates.io:

```
[dependencies]
linnet = "0.17"
```

Programmatic construction uses `HedgeGraphBuilder`. DOT support provides a convenient human-readable interchange and debugging format; it can carry global, node, edge, and half-edge attributes. Treat DOT as a representation boundary, not as a substitute for the typed graph invariants. Validate or handle parser errors before running algorithms.

The drawing feature adds graph drawing and layout support. Layout is separate from graph identity: coordinates and rendering attributes may change without changing the topology.

1.3 Related crates

One graph layer, several consumers

Spensio uses Linnet for tensor-network connectivity and adds tensor contraction semantics. GammaLoop uses the graph model in collider workflows and adds process generation, integration, and persistent state.

The Linnet [README](#) provides a compact crate summary; the Rust API reference covers individual types and methods.

2 Tutorial

In this tutorial, you will construct a half-edge graph in Rust, validate its invariants, find its one independent cycle, and print DOT. The builder returns Linnet's typed node and half-edge identifiers directly, so you can see where graph identity enters the workflow.

2.1 Prerequisites

Create a Rust binary project and add the current Linnet release:

```
cargo new linnet-first-graph
cd linnet-first-graph
cargo add linnet@0.17
```

No drawing feature or external layout program is needed to construct the graph and emit DOT.

2.2 Build a graph with a boundary

Replace `src/main.rs` with the following program:

```
use linnet::half_edge::{
    builder::HedgeGraphBuilder,
    involution::Flow,
    HedgeGraph,
};

fn main() {
    let mut builder = HedgeGraphBuilder::new();
    let a = builder.add_node("a");
    let b = builder.add_node("b");
    let c = builder.add_node("c");

    builder.add_edge(a, b, "ab", false);
    builder.add_edge(b, c, "bc", false);
    builder.add_edge(c, a, "ca", false);
    builder.add_external_edge(a, "incoming", false, Flow::Sink);
    builder.add_external_edge(c, "outgoing", false, Flow::Source);

    let graph: HedgeGraph<&str, &str> = builder.build();
    graph.check().expect("the half-edge involution is valid");

    let (basis, _) = graph.cycle_basis();
    assert_eq!(basis.len(), 1);
    println!("{}", graph.base_dot());
}
```

Run it with `cargo run`. Success means the assertion passes and stdout contains a DOT digraph with nodes a, b, and c, three paired internal edges, and two dangling boundary edges. The external edges do not add a cycle, so the triangle's cycle-basis rank remains one.

Keep the returned indexes with their graph

a, b, and c are compact `NodeIndex` values owned by this graph construction. Do not serialize their numeric representation as a durable identity or reuse them against another graph. Mutating operations return the mappings needed to relate old and new storage.

2.3 Make the first useful variation

Add a fourth internal edge between a and b, run the program again, and change the assertion to `basis.len() == 2`. This demonstrates the key distinction: parallel paired half-edges add an independent cycle, while dangling identity half-edges describe the graph boundary.

From here, use a subgraph view to run traversal, cuts, or contraction against only the region you intend to modify. Enable Linnet's drawing feature only when you need layout rather than plain DOT interchange.

2.4 Troubleshooting and next steps

- A type-inference error around `build()` means Rust cannot choose node storage or data types; keep the explicit `HedgeGraph<&str, &str>` annotation from the example.
- A failed `check()` points to an involution or storage invariant. Validate immediately after parsing or construction, before applying graph algorithms.
- If a traversal includes an unexpected dangling edge, check the selected subgraph and the `Flow` assigned to each external edge.
- Continue with the subgraph-and-algorithms manual, then consult the Rust API reference for the exact return mappings of mutating operations and additional builder methods.

3 Subgraphs, traversal, cuts, and drawing

Linnet algorithms operate on a graph together with an explicit subgraph view. This lets one algorithm work on the full graph, a selected region, or the boundary created by cutting paired half-edges. Choose the subgraph representation by operation: bit-backed sets are efficient for repeated set algebra, while borrowed or mapped views avoid copying when identity must remain attached to the owning graph.

3.1 Traversal and connectivity

A traversal needs a starting half-edge or node, the active subgraph, and a policy describing which adjacent half-edges may be crossed. Results retain typed indexes into the original graph. Breadth-first traversal is useful for distance layers and connectivity; depth-first traversal is useful for trees, back edges, and decompositions. A traversal tree is a derived view, not a new graph, so mutations must be coordinated with the indexes it contains.

Connected components, bridges, articulation structure, and biconnected regions all depend on the chosen subgraph. Run them on the full graph only when external half-edges and excluded nodes should participate in the answer.

3.2 Cycles and cuts

`cycle_basis` returns an independent basis and rank information. `all_cycles` expands that basis and can grow exponentially; prefer a basis when assigning loop momenta or computing a loop count. Symmetric differences combine cycle bitsets without inventing new graph identities.

Cut enumeration separates required left and right node sets and returns the left region, cut content, and right region. A cut edge is represented through its half-edges, preserving which side owns each endpoint. Validate

direction/flow after enumeration when a physical Cutkosky or tensor-network interpretation depends on orientation.

Complexity is part of the API choice

Enumerating every cycle or every admissible cut is inherently combinatorial. Filter the subgraph and endpoint sets first, and use basis or connectivity operations when enumeration is not required by the caller.

3.3 Mutation boundaries

Extraction copies a selected region, excision separates a graph along its boundary, sewing joins compatible dangling half-edges, and contraction identifies structure. These operations return mappings or replacement indexes where needed. Keep those results: a numeric Hedge or NodeIndex from before a storage-changing operation is not automatically a valid index into the resulting graph.

3.4 DOT and drawing

DOT parsing and emission are interchange/debugging boundaries. Attributes can carry graph, node, edge, and half-edge data, but the typed Rust invariants are reconstructed by the parser; handle parser errors before running algorithms. The drawing feature adds layout and rendering. Layout coordinates are presentation data and may vary without changing topology, subgraph membership, or edge flow.

Tensor contraction uses Spenso

Linnet determines connectivity and graph transformations. Tensor compatibility, contraction cost, and execution belong to Spenso even when a Spenso network delegates its graph algorithms to Linnet.

4 Rust and Python APIs

4.1 Rust package

The `linnet` package is organized into these principal areas:

- `half_edge` provides HedgeGraph, builders, the involution and edge data, subgraph types, traversal trees, and graph algorithms;
- `parser` provides DOT parsing and DOT-backed graph data;
- `drawing` provides rendering data and layout helpers;
- `permutation`, `tree`, and `union_find` provide supporting algorithms and stores.

The generic graph types separate edge, vertex, and half-edge payloads from the node-storage implementation. Start with the [curated Rust API](#), then follow its Rustdoc links for complete generic bounds and method signatures; choose a concrete storage type according to the workflow and mutation behavior you need.

Cargo features enable optional capabilities:

- `serde` adds serialization traits;
- `bincode` adds binary encoding support;
- `drawing` adds the drawing stack;
- `symbolica` adds Symbolica interoperability.

An item shown in an all-features API reference may not be available in a default build. Check its required feature and your `Cargo.toml` before using it.

4.2 Standalone Python distribution

Distribution and import have different names

Install the Python distribution `linnet-py`, then import `linnet_py`. It requires Python 3.10 or newer. It is a standalone extension and is not a `symbolica.community` module. Its package version is independent of the Rust `linnet` version. Record both versions when diagnosing compatibility between Rust and Python code.

The Python binding focuses on DOT-backed graphs and mirrors typed identifiers and subgraphs:

```
from linnet_py import DotGraphBuilder

builder = DotGraphBuilder()
left = builder.add_node()
right = builder.add_node()
builder.add_edge(left, right)
graph = builder.build()

print(graph.dot())
```

`DotGraph` can also parse one graph from a string or file, or a set of graphs from a string. The exported classes include `Hedge`, `NodeIndex`, `EdgeIndex`, `Flow`, `Orientation`, graph attribute wrappers, `HedgePair`, `Subgraph`, `Cycle`, `OrientedCut`, `TraversalTree`, `DotGraph`, and `DotGraphBuilder`.

Mutable statement wrappers expose mapping-like access to DOT attributes. Consult the structured [Python API](#) for their lifetime and write-through behavior, and do not assume that a copied mapping remains attached to its graph.

5 Releases and change history

The `Linnet` changelog records changes to the Rust crate. The standalone `linnet-py` distribution has its own version and does not currently have a separate changelog in this repository.

The changelog records user-visible graph, parser, drawing, and algorithm changes. In particular, users moving across releases should check index ordering, DOT serialization, subgraph behavior, and feature-gated integration changes rather than assuming a serialized graph or incidental ordering is stable.

5.1 Choosing the matching documentation

The latest channel may describe unreleased behavior. Documentation under `snapshots/<tag>` is fixed to a tagged repository revision. Choose a snapshot whose component information lists your Rust crate version, and check the `linnet-py` version separately when using the Python bindings.

Before upgrading

Review changes to index ordering, DOT serialization, subgraph behavior, and enabled features. Re-validate serialized graphs and any code that depends on traversal or iteration order.

API reference

The packages available in this version are listed below. Follow an item link for its complete language reference.

linnet

Linnet's supported half-edge graph API. Algorithms are grouped in the order used by the manual and full Rustdoc.

Half-edge graph

Stores graph topology as paired half-edges with independent edge, vertex, and half-edge payloads.

The main graph data structure, representing a graph using the half-edge (or doubly connected edge list - DCEL) principle.

A `HedgeGraph` stores nodes and edges (represented as pairs of half-edges). It is generic over the data associated with edges (`E`), nodes/vertices (`V`), and the specific storage strategy used for nodes (`S`).

The half-edge representation allows for efficient traversal of graph topology, such as iterating around faces (in planar embeddings), finding incident edges to a node, or quickly accessing an edge's opposite half-edge.

Type Parameters

- `E`: The type of data associated with each edge (or pair of half-edges).
- `V`: The type of data associated with each node (vertex).
- `S`: The node storage strategy, implementing the `NodeStorage` trait.

This determines how node data and their connectivity to half-edges are stored. Defaults to `DefaultNodeStore<V>` (feature-selected; `nodestore-vec` uses `nodestore::NodeStorageVec<V>`).

```
#[derive(Clone, Debug, PartialEq, Eq, PartialOrd, Ord, Hash)]
#[cfg_attr(feature = "serde", derive(serde::Serialize, serde::Deserialize))]
#[cfg_attr(feature = "bincode",
derive(bincode_trait_derive::Encode, bincode_trait_derive::Decode))]
#[doc =
" The main graph data structure, representing a graph using the half-edge"
#[doc = " (or doubly connected edge list - DCEL) principle." ] #[doc = " ]
#[doc =
" A HedgeGraph stores nodes and edges (represented as pairs of half-edges)."]
#[doc =
" It is generic over the data associated with edges (E), nodes/vertices (V),"
#[doc = " and the specific storage strategy used for nodes (S)."]
#[doc = " ]
#[doc =
" The half-edge representation allows for efficient traversal of graph topology,"
#[doc =
" such as iterating around faces (in planar embeddings), finding incident edges"
#[doc = " to a node, or quickly accessing an edge's opposite half-edge." ]
#[doc = " ] #[doc = " # Type Parameters" ] #[doc = " ]
#[doc =
" - E: The type of data associated with each edge (or pair of half-edges)."]
#[doc = " - V: The type of data associated with each node (vertex)."]
#[doc =
" - S: The node storage strategy, implementing the NodeStorage trait." ]
```

```

#[doc =
"   This determines how node data and their connectivity to half-edges"]
#[doc =
"   are stored. Defaults to [DefaultNodeStore<V>] (feature-selected; nodestore-vec` uses")
#[doc = "   [nodestore::NodeStorageVec<V>])."] pub struct HedgeGraph < E, V,
H = NoData, S : NodeStorage < NodeData = V > = DefaultNodeStore < V > >
{
    hedge_data : HedgeVec < H > ,
    #[doc =
    "   Internal storage for all half-edges, their data, and their topological"
    #[doc =
    "   relationships (e.g., opposite half-edge, next half-edge around a node)."]
    #[doc = "   This is typically a [SmartHedgeVec<E>]."] edge_store :
    SmartEdgeVec < E > ,
    #[doc =
    "   Storage for all nodes in the graph, including their data (V`) and"
    #[doc = "   information about the half-edges incident to them."]
    #[doc =
    "   The specific implementation is determined by the S` type parameter."]
    pub node_store : S,
}

```

Reference feature matrix: `nodestore-vec`, `serde`, `bincode`, `drawing`, `symbolica` (item is available without an additional gate)

Supported usage

```
use linnet::half_edge::HedgeGraph;
```

hedge_data

Kind: Field

HedgeVec < H >

edge_store

Kind: Field

SmartEdgeVec < E >

Internal storage for all half-edges, their data, and their topological relationships (e.g., opposite half-edge, next half-edge around a node). This is typically a [`SmartHedgeVec<E>`].

node_store

Kind: Field

S

Storage for all nodes in the graph, including their data (`V``) and information about the half-edges incident to them. The specific implementation is determined by the `S`` type parameter.

Exhaustive reference · Source

Graph builder

Builds a valid half-edge graph while assigning typed node, edge, and hedge identifiers.

A builder for programmatically constructing [`HedgeGraph``] instances.

This builder allows for the incremental addition of nodes and edges (both paired and external/dangling) before finalizing the graph structure.

Type Parameters

- `E`: The type of data to be associated with edges.
- `V`: The type of data to be associated with nodes.

Example

```
```rust,ignore
use linnet::half_edge::builder::HedgeGraphBuilder;
use linnet::half_edge::involution::{Flow, Orientation};

let mut builder = HedgeGraphBuilder::<&str, &str>::new();

let node0 = builder.add_node("Node_0_Data");
let node1 = builder.add_node("Node_1_Data");
let node2 = builder.add_node("Node_2_Data");

builder.add_edge(node0, node1, "Edge_01_Data", Orientation::Default);
builder.add_external_edge(node2, "External_Edge_Data", Orientation::Undirected,
Flow::Source);

// Assuming a NodeStorage type MyNodeStore is defined and implements NodeStorageOps
// let graph: HedgeGraph<&str, &str, MyNodeStore> = builder.build();
```

#[derive(Clone, Debug)]
#[doc =
" A builder for programmatically constructing [HedgeGraph] instances."]
#[doc = ""]
#[doc =
" This builder allows for the incremental addition of nodes and edges (both"
#[doc =
" paired and external/dangling) before finalizing the graph structure."]
#[doc = ""] #[doc = " # Type Parameters"] #[doc = ""]
#[doc = " - `E`: The type of data to be associated with edges."]
#[doc = " - `V`: The type of data to be associated with nodes."] #[doc = ""]
#[doc = " # Example"] #[doc = ""] #[doc = " ```rust,ignore"]
#[doc = " use linnet::half_edge::builder::HedgeGraphBuilder;"
#[doc = " use linnet::half_edge::involution::{Flow, Orientation};"]
#[doc = ""]
#[doc = " let mut builder = HedgeGraphBuilder::<&str, &str>::new();"]
#[doc = ""] #[doc = " let node0 = builder.add_node(\"Node_0_Data\");"]
#[doc = " let node1 = builder.add_node(\"Node_1_Data\");"]
#[doc = " let node2 = builder.add_node(\"Node_2_Data\");"] #[doc = ""]
#[doc =
" builder.add_edge(node0, node1, \"Edge_01_Data\", Orientation::Default);"]
#[doc =
" builder.add_external_edge(node2, \"External_Edge_Data\", Orientation::Undirected,
Flow::Source);"]
#[doc = ""]
#[doc =
" // Assuming a NodeStorage type MyNodeStore is defined and implements
NodeStorageOps"]
#[doc =
```

```
" // let graph: HedgeGraph<&str, &str, MyNodeStore> = builder.build();"
#[doc = " ``"] pub struct HedgeGraphBuilder < E, V, H = NoData >
{
    hedge_data : HedgeVec < H > ,
    #[doc =
        " A list of nodes currently being built, stored as [`HedgeNodeBuilder`]
instances."]
    nodes : Vec < HedgeNodeBuilder < V > > ,
    #[doc =
        " The [`Involution`] structure managing the half-edges being added to the
graph."]
    pub(crate) involution : Involution < E > ,
}
```

Reference feature matrix: `nodestore-vec`, `serde`, `bincode`, `drawing`, `symbolica` (item is available without an additional gate)

Example 1

```
use linnet::half_edge::builder::HedgeGraphBuilder;
use linnet::half_edge::involution::{Flow, Orientation};

let mut builder = HedgeGraphBuilder::<&str, &str>::new();

let node0 = builder.add_node("Node_0_Data");
let node1 = builder.add_node("Node_1_Data");
let node2 = builder.add_node("Node_2_Data");

builder.add_edge(node0, node1, "Edge_01_Data", Orientation::Default);
builder.add_external_edge(node2, "External_Edge_Data", Orientation::Undirected,
Flow::Source);

// Assuming a NodeStorage type MyNodeStore is defined and implements NodeStorageOps
// let graph: HedgeGraph<&str, &str, MyNodeStore> = builder.build();
```

Supported usage

```
use linnet::half_edge::builder::HedgeGraphBuilder;
```

hedge_data

Kind: Field

HedgeVec < H >

nodes

Kind: Field

Vec < HedgeNodeBuilder < V > >

A list of nodes currently being built, stored as [`HedgeNodeBuilder`] instances.

involution

Kind: Field

Involution < E >

The [`Involution`] structure managing the half-edges being added to the graph.

Exhaustive reference · Source

Bit-vector subgraph

Represents a subgraph as an efficient half-edge inclusion filter.

Represents a subgraph as an efficient half-edge inclusion filter.

```
pub type SuBitGraph = SubSet < Hedge > ;
```

Reference feature matrix: nodestore-vec, serde, bincode, drawing, symbolica (item is available without an additional gate)

Supported usage

```
use linnet::half_edge::subgraph::SuBitGraph;
```

Exhaustive reference · Source

Traversal tree

Records parent, root, and traversal information produced by breadth-first or depth-first graph walks.

Represents a traversal tree (e.g., DFS or BFS tree) derived from a [`HedgeGraph``].

This structure uses a generic [`Forest``] to store the tree topology. Each "root" in the

`Forest`` corresponds to a node in the original `HedgeGraph``, and its data is a [`TTRoot``]

enum indicating its role (Root, Child, or None) in this specific traversal. The nodes within each tree of the `Forest`` typically represent the half-edges that form the traversal path.

Type Parameters

- `P``: The type of [`ForestNodeStore``] used by the underlying `Forest``.

It stores `()`` as `NodeData`` because the tree nodes (representing half-edges) don't need additional data beyond their structural role in the traversal tree.

Defaults to [`ParentPointerStore<()>``].

```
#[derive(Debug, Clone)]
```

```
#[cfg_attr(feature = "serde", derive(serde::Serialize, serde::Deserialize))]
```

```
#[doc =
```

```
" Represents a traversal tree (e.g., DFS or BFS tree) derived from a  
[`HedgeGraph`]."]
```

```
#[doc = ""]
```

```
#[doc =
```

```
" This structure uses a generic [`Forest`] to store the tree topology. Each \"root\"  
in the"]
```

```
#[doc =
```

```
" `Forest` corresponds to a node in the original `HedgeGraph`, and its data is a  
[`TTRoot`]"
```

```
#[doc =
```

```
" enum indicating its role (Root, Child, or None) in this specific traversal. The  
nodes"]
```

```
#[doc =
```

```
" within each tree of the `Forest` typically represent the half-edges that form the"  
#[doc = " traversal path." ] #[doc = ""] #[doc = " # Type Parameters"]
```

```
#[doc = ""]
```

```
#[doc =
```

```
" - P`: The type of [ForestNodeStore`] used by the underlying Forest`." ]
```

```
#[doc =
```

```
" It stores ()` as NodeData` because the tree nodes (representing half-edges)" ]
```

```

#[doc =
" don't need additional data beyond their structural role in the traversal tree."]
#[doc = " Defaults to [ParentPointerStore<()>]."] pub struct
SimpleTraversalTree < P : ForestNodeStore < NodeData = () > =
ParentPointerStore < () > >
{
    #[doc =
    " The underlying forest structure representing the traversal tree(s)."]
    #[doc =
    " Each root in this forest is a graph node, and its data is its [TTRoot`]
status." ]
    #[doc = " Nodes within each tree of this forest represent half-edges." ]
    forest : Forest < TTRoot, P > ,
    #[doc =
    " A bitmask representing the set of half-edges that constitute the actual" ]
    #[doc = " edges of this traversal tree." ] pub tree_subgraph : SuBitGraph,
}

```

Reference feature matrix: nodestore-vec, serde, bincode, drawing, symbolica (item is available without an additional gate)

Supported usage

```
use linnet::half_edge::tree::SimpleTraversalTree;
```

forest

Kind: Field

Forest < TTRoot, P >

The underlying forest structure representing the traversal tree(s).
Each root in this forest is a graph node, and its data is its [`TTRoot``] status.
Nodes within each tree of this forest represent half-edges.

tree_subgraph

Kind: Field

SuBitGraph

A bitmask representing the set of half-edges that constitute the actual
edges of this traversal tree.

Exhaustive reference · Source

DOT-backed graph

Combines a half-edge graph with DOT attributes and parser/serializer support.

Combines a half-edge graph with DOT attributes and parser/serializer support.

```

#[derive(Debug, Clone, PartialEq, Eq)] pub struct DotGraph < N : NodeStorage <
NodeData = DotVertexData > = DefaultNodeStore < DotVertexData > >
{
    pub global_data : GlobalData, pub graph : HedgeGraph < DotEdgeData,
    DotVertexData, DotHedgeData, N > ,
}

```

Reference feature matrix: nodestore-vec, serde, bincode, drawing, symbolica (item is available without an additional gate)

Supported usage

```
use linnet::parser::DotGraph;
```

global_data

Kind: Field

GlobalData

graph

Kind: Field

HedgeGraph < DotEdgeData, DotVertexData, DotHedgeData, N >

[Exhaustive reference](#) · [Source](#)

linnet-py

Exports registered by linnet-py.

Cycle

Cycle represented as a subgraph and optional loop count.

Cycle represented as a subgraph and optional loop count.

```
class Cycle:
```

Requires features: extension-module, abi3-py310

Inspect the registered export

```
from linnet_py import Cycle
help(Cycle)
```

filter

Kind: Getter

```
def filter(self) -> Subgraph:
```

Subgraph filter for this cycle.

loop_count

Kind: Getter

```
def loop_count(self) -> typing.Optional[builtins.int]:
```

Optional loop count for this cycle.

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

DotEdgeData

DOT edge data (attributes + optional edge id).

DOT edge data (attributes + optional edge id).

```
class DotEdgeData:
```

Requires features: extension-module, abi3-py310

Inspect the registered export

```
from linnet_py import DotEdgeData
help(DotEdgeData)
```

statements**Kind:** Getter

```
def statements(self) -> EdgeStatements:  
Edge attributes as a dict-like proxy.
```

local_statements**Kind:** Getter

```
def local_statements(self) -> EdgeStatements:  
Local edge attributes as a dict-like proxy.
```

edge_id**Kind:** Getter

```
def edge_id(self) -> typing.Optional[EdgeIndex]:  
Optional edge id.
```

__new__**Kind:** AssociatedFunction

```
def __new__(cls, statements: typing.Optional[typing.Mapping[builtins.str,  
builtins.str]] = None, local_statements: typing.Optional[typing.Mapping[builtins.str,  
builtins.str]] = None, edge_id: typing.Optional[builtins.int] = None) -> DotEdgeData:  
Create edge data from fields.
```

statements**Kind:** Parameter

```
statements: typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None
```

Default: None**local_statements****Kind:** Parameter

```
local_statements: typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None
```

Default: None**edge_id****Kind:** Parameter

```
edge_id: typing.Optional[builtins.int] = None
```

Default: None**add_statement****Kind:** Method

```
def add_statement(self, key: builtins.str, value: builtins.str) -> None:  
Insert or update an attribute statement.
```

key**Kind:** Parameter

```
key: builtins.str
```

value

Kind: Parameter

value: builtins.str

__repr__

Kind: Method

def __repr__(self) -> builtins.str:

Debug-style representation.

Exhaustive reference · Source

DotGraph

DOT-backed hedge graph.

DOT-backed hedge graph.

class DotGraph:

Requires features: extension-module, abi3-py310

Inspect the registered export

```
from linnet_py import DotGraph
help(DotGraph)
```

global_data

Kind: Getter

def global_data(self) -> GlobalData:

Global graph attributes.

global_data

Kind: Setter

def global_data(self, value: GlobalData) -> None:

value

Kind: Parameter

value: GlobalData

from_string

Kind: AssociatedFunction

def from_string(cls, s: builtins.str) -> DotGraph:

Parse a DOT string into a graph.

s

Kind: Parameter

s: builtins.str

from_string_set

Kind: AssociatedFunction

def from_string_set(cls, s: builtins.str) -> builtins.list[DotGraph]:

Parse a DOT string into multiple graphs.

s

Kind: Parameter

s: builtins.str

from_file

Kind: AssociatedFunction

def from_file(cls, path: builtins.str) -> DotGraph:

Parse a DOT file into a graph.

path

Kind: Parameter

path: builtins.str

debug_dot

Kind: Method

def debug_dot(self) -> builtins.str:

Serialize graph to DOT for debugging.

dot

Kind: Method

def dot(self) -> builtins.str:

Serialize the full graph to DOT.

dot_of

Kind: Method

def dot_of(self, subgraph: Subgraph) -> builtins.str:

Serialize a subgraph to DOT.

subgraph

Kind: Parameter

subgraph: Subgraph

n_nodes

Kind: Method

def n_nodes(self) -> builtins.int:

Number of nodes.

n_edges

Kind: Method

def n_edges(self) -> builtins.int:

Number of edges.

n_hedges

Kind: Method

def n_hedges(self) -> builtins.int:

Number of hedges.

n_externals**Kind:** Method

```
def n_externals(self) -> builtins.int:  
    Number of external hedges.
```

n_internals**Kind:** Method

```
def n_internals(self) -> builtins.int:  
    Number of internal hedges.
```

full_filter**Kind:** Method

```
def full_filter(self) -> Subgraph:  
    Subgraph including all hedges.
```

empty_subgraph**Kind:** Method

```
def empty_subgraph(self) -> Subgraph:  
    Empty subgraph of this graph.
```

iter_edges**Kind:** Method

```
def iter_edges(self) -> builtins.list[tuple[HedgePair, EdgeIndex, EdgeData]]:  
    Iterate edges in the full graph.
```

iter_edges_of**Kind:** Method

```
def iter_edges_of(self, subgraph: Subgraph) -> builtins.list[tuple[HedgePair,  
    EdgeIndex, EdgeData]]:  
    Iterate edges within a subgraph.
```

subgraph**Kind:** Parameter

subgraph: Subgraph

iter_nodes**Kind:** Method

```
def iter_nodes(self) -> builtins.list[tuple[NodeIndex, builtins.list[Hedge],  
    DotVertexData]]:  
    Iterate nodes in the full graph.
```

iter_nodes_of**Kind:** Method

```
def iter_nodes_of(self, subgraph: Subgraph) -> builtins.list[tuple[NodeIndex,  
    builtins.list[Hedge], DotVertexData]]:  
    Iterate nodes within a subgraph.
```

subgraph**Kind:** Parameter

subgraph: Subgraph

connected_components**Kind:** Method

```
def connected_components(self, subgraph: Subgraph) -> builtins.list[Subgraph]:
```

Connected components of a subgraph.

subgraph**Kind:** Parameter

subgraph: Subgraph

count_connected_components**Kind:** Method

```
def count_connected_components(self, subgraph: Subgraph) -> builtins.int:
```

Count connected components.

subgraph**Kind:** Parameter

subgraph: Subgraph

is_connected**Kind:** Method

```
def is_connected(self, subgraph: Subgraph) -> builtins.bool:
```

Whether a subgraph is connected.

subgraph**Kind:** Parameter

subgraph: Subgraph

depth_first_traverse**Kind:** Method

```
def depth_first_traverse(self, subgraph: Subgraph, root_node: NodeIndex,  
include_hedge: typing.Optional[Hedge]) -> TraversalTree:
```

Depth-first traversal from a root node.

subgraph**Kind:** Parameter

subgraph: Subgraph

root_node**Kind:** Parameter

root_node: NodeIndex

include_hedge**Kind:** Parameter

include_hedge: typing.Optional[Hedge]

breadth_first_traverse

Kind: Method

```
def breadth_first_traverse(self, subgraph: Subgraph, root_node: NodeIndex,  
include_hedge: typing.Optional[Hedge]) -> TraversalTree:
```

Breadth-first traversal from a root node.

subgraph

Kind: Parameter

subgraph: Subgraph

root_node

Kind: Parameter

root_node: NodeIndex

include_hedge

Kind: Parameter

include_hedge: typing.Optional[Hedge]

bridges

Kind: Method

```
def bridges(self) -> Subgraph:
```

Bridges in the full graph.

bridges_of

Kind: Method

```
def bridges_of(self, subgraph: Subgraph) -> Subgraph:
```

Bridges within a subgraph.

subgraph

Kind: Parameter

subgraph: Subgraph

cycle_basis

Kind: Method

```
def cycle_basis(self) -> tuple[builtins.list[Cycle], Subgraph]:
```

Cycle basis of the full graph.

cycle_basis_of

Kind: Method

```
def cycle_basis_of(self, subgraph: Subgraph) -> tuple[builtins.list[Cycle],  
Subgraph]:
```

Cycle basis of a subgraph.

subgraph

Kind: Parameter

subgraph: Subgraph

all_spanning_forests

Kind: Method

```
def all_spanning_forests(self) -> builtins.list[Subgraph]:
```

All spanning forests of the full graph.

all_spanning_forests_of

Kind: Method

```
def all_spanning_forests_of(self, subgraph: Subgraph) -> builtins.list[Subgraph]:
```

All spanning forests of a subgraph.

subgraph

Kind: Parameter

subgraph: Subgraph

combine_to_single_hedgenode

Kind: Method

```
def combine_to_single_hedgenode(self, nodes: typing.Sequence[NodeIndex]) -> HedgeNode:
```

Combine nodes into a single hedge node.

nodes

Kind: Parameter

nodes: typing.Sequence[NodeIndex]

all_cuts

Kind: Method

```
def all_cuts(self, source: HedgeNode, target: HedgeNode) -> builtins.list[tuple[Subgraph, OrientedCut, Subgraph]]:
```

All cuts between two hedge nodes.

source

Kind: Parameter

source: HedgeNode

target

Kind: Parameter

target: HedgeNode

all_cuts_from_ids

Kind: Method

```
def all_cuts_from_ids(self, source: typing.Sequence[NodeIndex], target: typing.Sequence[NodeIndex]) -> builtins.list[tuple[Subgraph, OrientedCut, Subgraph]]:
```

All cuts between two sets of node indices.

source

Kind: Parameter

source: typing.Sequence[NodeIndex]

target**Kind:** Parameter

target: typing.Sequence[NodeIndex]

contract_subgraph**Kind:** Method

```
def contract_subgraph(self, subgraph: Subgraph, node_data_merge:
typing.Optional[DotVertexData] = None) -> None:
```

Contract a subgraph into a single node, deleting its edges.

subgraph**Kind:** Parameter

subgraph: Subgraph

node_data_merge**Kind:** Parameter

node_data_merge: typing.Optional[DotVertexData] = None

Default: None**join****Kind:** Method

```
def join(self, other: DotGraph, matching_fn: typing.Any, merge_fn: typing.Any) ->
DotGraph:
```

Join two graphs, matching dangling edges via a Python callback.

other**Kind:** Parameter

other: DotGraph

matching_fn**Kind:** Parameter

matching_fn: typing.Any

merge_fn**Kind:** Parameter

merge_fn: typing.Any

join_mut**Kind:** Method

```
def join_mut(self, other: DotGraph, matching_fn: typing.Any, merge_fn: typing.Any) ->
None:
```

In-place join, matching dangling edges via a Python callback.

other**Kind:** Parameter

other: DotGraph

matching_fn

Kind: Parameter

matching_fn: typing.Any

merge_fn

Kind: Parameter

merge_fn: typing.Any

extract

Kind: Method

```
def extract(self, subgraph: Subgraph, split_edge_fn: typing.Any, internal_data:
typing.Any, split_node: typing.Any, owned_node: typing.Any) -> DotGraph:
```

Extract a subgraph with Python callbacks to transform edge/node data.

subgraph

Kind: Parameter

subgraph: Subgraph

split_edge_fn

Kind: Parameter

split_edge_fn: typing.Any

internal_data

Kind: Parameter

internal_data: typing.Any

split_node

Kind: Parameter

split_node: typing.Any

owned_node

Kind: Parameter

owned_node: typing.Any

__getitem__

Kind: Method

```
def __getitem__(self, key: Hedge) -> DotHedgeData:
```

key

Kind: Parameter

key: Hedge

__getitem__

Kind: Method

```
def __getitem__(self, key: NodeIndex) -> DotVertexData:
```

key

Kind: Parameter

key: NodeIndex

__getitem__

Kind: Method

```
def __getitem__(self, key: EdgeIndex) -> DotEdgeData:
```

key

Kind: Parameter

key: EdgeIndex

Exhaustive reference · Source

DotGraphBuilder

Builder for constructing DOT graphs programmatically.

Builder for constructing DOT graphs programmatically.

```
class DotGraphBuilder:
```

Requires features: extension-module, abi3-py310

Inspect the registered export

```
from linnet_py import DotGraphBuilder
help(DotGraphBuilder)
```

__new__

Kind: AssociatedFunction

```
def __new__(cls) -> DotGraphBuilder:
```

Create a new, empty graph builder.

add_node

Kind: Method

```
def add_node(self, data: typing.Optional[DotVertexData] = None) -> NodeIndex:
```

Add a node and return its index.

data

Kind: Parameter

data: typing.Optional[DotVertexData] = None

Default: None

add_edge

Kind: Method

```
def add_edge(self, source: NodeIndex, sink: NodeIndex, data:
typing.Optional[DotEdgeData] = None, orientation: typing.Optional[Orientation] =
None, source_hedge: typing.Optional[DotHedgeData] = None, sink_hedge:
typing.Optional[DotHedgeData] = None) -> None:
```

Add an edge between two nodes.

source

Kind: Parameter

source: NodeIndex

sink**Kind:** Parameter

sink: NodeIndex

data**Kind:** Parameter

data: typing.Optional[DotEdgeData] = None

Default: None**orientation****Kind:** Parameter

orientation: typing.Optional[Orientation] = None

Default: None**source_hedge****Kind:** Parameter

source_hedge: typing.Optional[DotHedgeData] = None

Default: None**sink_hedge****Kind:** Parameter

sink_hedge: typing.Optional[DotHedgeData] = None

Default: None**add_external_edge****Kind:** Method

```
def add_external_edge(self, source: NodeIndex, data: typing.Optional[DotEdgeData] = None, orientation: typing.Optional[Orientation] = None, flow: typing.Optional[Flow] = None, hedge: typing.Optional[DotHedgeData] = None) -> None:
```

Add a dangling (external) edge incident to a node.

source**Kind:** Parameter

source: NodeIndex

data**Kind:** Parameter

data: typing.Optional[DotEdgeData] = None

Default: None**orientation****Kind:** Parameter

orientation: typing.Optional[Orientation] = None

Default: None

flow

Kind: Parameter

flow: typing.Optional[Flow] = None

Default: None

hedge

Kind: Parameter

hedge: typing.Optional[DotHedgeData] = None

Default: None

build

Kind: Method

```
def build(self) -> DotGraph:
```

Build the graph and reset the builder.

Exhaustive reference · Source

DotHedgeData

DOT hedge (half-edge) data.

DOT hedge (half-edge) data.

```
class DotHedgeData:
```

Requires features: extension-module, abi3-py310

Inspect the registered export

```
from linnet_py import DotHedgeData
```

```
help(DotHedgeData)
```

statement

Kind: Getter

```
def statement(self) -> typing.Optional[builtins.str]:
```

Optional statement string.

id

Kind: Getter

```
def id(self) -> typing.Optional[Hedge]:
```

Optional hedge id.

port_label

Kind: Getter

```
def port_label(self) -> typing.Optional[builtins.str]:
```

Optional port label.

compasspt

Kind: Getter

```
def compasspt(self) -> typing.Optional[builtins.str]:
```

Optional compass point as a string.

`__new__`

Kind: AssociatedFunction

```
def __new__(cls, statement: typing.Optional[builtins.str] = None, id:
typing.Optional[builtins.int] = None, port_label: typing.Optional[builtins.str] =
None, compasspt: typing.Optional[builtins.str] = None) -> DotHedgeData:
```

Create hedge data from fields.

statement

Kind: Parameter

```
statement: typing.Optional[builtins.str] = None
```

Default: None

id

Kind: Parameter

```
id: typing.Optional[builtins.int] = None
```

Default: None

port_label

Kind: Parameter

```
port_label: typing.Optional[builtins.str] = None
```

Default: None

compasspt

Kind: Parameter

```
compasspt: typing.Optional[builtins.str] = None
```

Default: None

`__repr__`

Kind: Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

Exhaustive reference · Source

DotVertexData

DOT vertex data (name, optional index, and attribute statements).

DOT vertex data (name, optional index, and attribute statements).

```
class DotVertexData:
```

Requires features: extension-module, abi3-py310

Inspect the registered export

```
from linnet_py import DotVertexData
help(DotVertexData)
```

name

Kind: Getter

```
def name(self) -> typing.Optional[builtins.str]:
```

Optional vertex name.

index

Kind: Getter

```
def index(self) -> typing.Optional[NodeIndex]:
```

Optional node index.

statements

Kind: Getter

```
def statements(self) -> NodeStatements:
```

Attribute statements as a dict-like proxy.

__new__

Kind: AssociatedFunction

```
def __new__(cls, name: typing.Optional[builtins.str] = None, index:
typing.Optional[builtins.int] = None, statements:
typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None) -> DotVertexData:
Create vertex data from fields.
```

name

Kind: Parameter

```
name: typing.Optional[builtins.str] = None
```

Default: None

index

Kind: Parameter

```
index: typing.Optional[builtins.int] = None
```

Default: None

statements

Kind: Parameter

```
statements: typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None
```

Default: None

add_statement

Kind: Method

```
def add_statement(self, key: builtins.str, value: builtins.str) -> None:
```

Insert or update an attribute statement.

key

Kind: Parameter

```
key: builtins.str
```

value

Kind: Parameter

```
value: builtins.str
```

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

Exhaustive reference · Source

EdgeData

Edge data with orientation.

Edge data with orientation.

```
class EdgeData:
```

Requires features: extension-module, abi3-py310

Inspect the registered export

```
from linnet_py import EdgeData
help(EdgeData)
```

orientation

Kind: Getter

```
def orientation(self) -> Orientation:
```

Orientation of this edge.

data

Kind: Getter

```
def data(self) -> DotEdgeData:
```

DOT edge data.

__new__

Kind: AssociatedFunction

```
def __new__(cls, orientation: typing.Any, data: typing.Any) -> EdgeData:
```

Create edge data from orientation and DotEdgeData.

orientation

Kind: Parameter

orientation: typing.Any

data

Kind: Parameter

data: typing.Any

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

Exhaustive reference · Source

EdgeIndex

Edge index identifier.

Edge index identifier.

class EdgeIndex:

Requires features: extension-module, abi3-py310

Inspect the registered export

```
from linnet_py import EdgeIndex
help(EdgeIndex)
```

value

Kind: Getter

```
def value(self) -> builtins.int:
```

Numeric value of this edge index.

__new__

Kind: AssociatedFunction

```
def __new__(cls, value: builtins.int) -> EdgeIndex:
```

Create an edge index from a zero-based index.

value

Kind: Parameter

value: builtins.int

__int__

Kind: Method

```
def __int__(self) -> builtins.int:
```

Convert to int.

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

[Exhaustive reference · Source](#)

EdgeStatements

Dict-like proxy for edge attribute statements.

Dict-like proxy for edge attribute statements.

class EdgeStatements:

Requires features: extension-module, abi3-py310

Inspect the registered export

```
from linnet_py import EdgeStatements
help(EdgeStatements)
```

__getitem__

Kind: Method

```
def __getitem__(self, key: builtins.str) -> builtins.str:
```

key

Kind: Parameter

key: builtins.str

__setitem__

Kind: Method

def __setitem__(self, key: builtins.str, value: builtins.str) -> None:

key

Kind: Parameter

key: builtins.str

value

Kind: Parameter

value: builtins.str

__delitem__

Kind: Method

def __delitem__(self, key: builtins.str) -> None:

key

Kind: Parameter

key: builtins.str

__contains__

Kind: Method

def __contains__(self, key: builtins.str) -> builtins.bool:

key

Kind: Parameter

key: builtins.str

__len__

Kind: Method

def __len__(self) -> builtins.int:

__iter__

Kind: Method

def __iter__(self) -> typing.Iterator[builtins.str]:

get

Kind: Method

def get(self, key: builtins.str, default: typing.Any = None) -> typing.Any:

key

Kind: Parameter

key: builtins.str

default**Kind:** Parameter

default: typing.Any = None

Default: None**keys****Kind:** Method

def keys(self) -> builtins.list[builtins.str]:

values**Kind:** Method

def values(self) -> builtins.list[builtins.str]:

items**Kind:** Method

def items(self) -> builtins.list[tuple[builtins.str, builtins.str]]:

clear**Kind:** Method

def clear(self) -> None:

update**Kind:** Method

def update(self, other: typing.Mapping[builtins.str, builtins.str]) -> None:

other**Kind:** Parameter

other: typing.Mapping[builtins.str, builtins.str]

pop**Kind:** Method

def pop(self, key: builtins.str, default: typing.Any = None) -> typing.Any:

key**Kind:** Parameter

key: builtins.str

default**Kind:** Parameter

default: typing.Any = None

Default: None**popitem****Kind:** Method

def popitem(self) -> tuple[builtins.str, builtins.str]:

setdefault**Kind:** Method

```
def setdefault(self, key: builtins.str, default: typing.Optional[builtins.str] =
None) -> builtins.str:
```

key

Kind: Parameter

key: builtins.str

default

Kind: Parameter

default: typing.Optional[builtins.str] = None

Default: None

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:
```

Exhaustive reference · Source

Flow

Flow direction (Source/Sink).

Flow direction (Source/Sink).

```
class Flow:
```

Requires features: extension-module, abi3-py310

Inspect the registered export

```
from linnet_py import Flow
help(Flow)
```

source

Kind: AssociatedFunction

```
def source() -> Flow:
```

Flow::Source.

sink

Kind: AssociatedFunction

```
def sink() -> Flow:
```

Flow::Sink.

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

Exhaustive reference · Source

GlobalData

Global graph attributes (graph/node/edge statements).

Global graph attributes (graph/node/edge statements).

```
class GlobalData:
```

Requires features: extension-module, abi3-py310

Inspect the registered export

```
from linnet_py import GlobalData
help(GlobalData)
```

name

Kind: Getter

```
def name(self) -> builtins.str:
```

Graph name.

name

Kind: Setter

```
def name(self, value: builtins.str) -> None:
```

value

Kind: Parameter

```
value: builtins.str
```

statements

Kind: Getter

```
def statements(self) -> GlobalStatements:
```

Graph-level statements.

statements

Kind: Setter

```
def statements(self, value: builtins.dict[builtins.str, builtins.str]) -> None:
```

value

Kind: Parameter

```
value: builtins.dict[builtins.str, builtins.str]
```

edge_statements

Kind: Getter

```
def edge_statements(self) -> GlobalStatements:
```

Default edge statements.

edge_statements

Kind: Setter

```
def edge_statements(self, value: builtins.dict[builtins.str, builtins.str]) -> None:
```

value

Kind: Parameter

```
value: builtins.dict[builtins.str, builtins.str]
```

node_statements

Kind: Getter

```
def node_statements(self) -> GlobalStatements:
```

Default node statements.

node_statements

Kind: Setter

```
def node_statements(self, value: builtins.dict[builtins.str, builtins.str]) -> None:
```

value

Kind: Parameter

```
value: builtins.dict[builtins.str, builtins.str]
```

__new__

Kind: AssociatedFunction

```
def __new__(cls, name: typing.Optional[builtins.str] = None, statements:
typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None, edge_statements:
typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None, node_statements:
typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None) -> GlobalData:
```

Create global data from fields.

name

Kind: Parameter

```
name: typing.Optional[builtins.str] = None
```

Default: None

statements

Kind: Parameter

```
statements: typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None
```

Default: None

edge_statements

Kind: Parameter

```
edge_statements: typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None
```

Default: None

node_statements

Kind: Parameter

```
node_statements: typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None
```

Default: None

add_statement

Kind: Method

```
def add_statement(self, key: builtins.str, value: builtins.str) -> None:
```

Insert or update a graph-level statement.

key

Kind: Parameter

```
key: builtins.str
```

value

Kind: Parameter

value: builtins.str

add_edge_statement

Kind: Method

```
def add_edge_statement(self, key: builtins.str, value: builtins.str) -> None:
```

Insert or update a default edge statement.

key

Kind: Parameter

key: builtins.str

value

Kind: Parameter

value: builtins.str

add_node_statement

Kind: Method

```
def add_node_statement(self, key: builtins.str, value: builtins.str) -> None:
```

Insert or update a default node statement.

key

Kind: Parameter

key: builtins.str

value

Kind: Parameter

value: builtins.str

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

GlobalStatements

Dict-like proxy for global statements.

Dict-like proxy for global statements.

```
class GlobalStatements:
```

Requires features: extension-module, abi3-py310

Inspect the registered export

```
from linnet_py import GlobalStatements
help(GlobalStatements)
```

__getitem__

Kind: Method

```
def __getitem__(self, key: builtins.str) -> builtins.str:
```

key

Kind: Parameter

key: builtins.str

__setitem__

Kind: Method

```
def __setitem__(self, key: builtins.str, value: builtins.str) -> None:
```

key

Kind: Parameter

key: builtins.str

value

Kind: Parameter

value: builtins.str

__delitem__

Kind: Method

```
def __delitem__(self, key: builtins.str) -> None:
```

key

Kind: Parameter

key: builtins.str

__contains__

Kind: Method

```
def __contains__(self, key: builtins.str) -> builtins.bool:
```

key

Kind: Parameter

key: builtins.str

__len__

Kind: Method

```
def __len__(self) -> builtins.int:
```

__iter__

Kind: Method

```
def __iter__(self) -> typing.Iterator[builtins.str]:
```

get

Kind: Method

```
def get(self, key: builtins.str, default: typing.Any = None) -> typing.Any:
```

key**Kind:** Parameter

key: builtins.str

default**Kind:** Parameter

default: typing.Any = None

Default: None**keys****Kind:** Method

def keys(self) -> builtins.list[builtins.str]:

values**Kind:** Method

def values(self) -> builtins.list[builtins.str]:

items**Kind:** Method

def items(self) -> builtins.list[tuple[builtins.str, builtins.str]]:

clear**Kind:** Method

def clear(self) -> None:

update**Kind:** Method

def update(self, other: typing.Mapping[builtins.str, builtins.str]) -> None:

other**Kind:** Parameter

other: typing.Mapping[builtins.str, builtins.str]

pop**Kind:** Method

def pop(self, key: builtins.str, default: typing.Any = None) -> typing.Any:

key**Kind:** Parameter

key: builtins.str

default**Kind:** Parameter

default: typing.Any = None

Default: None**popitem****Kind:** Method

```
def popitem(self) -> tuple[builtins.str, builtins.str]:
```

setdefault

Kind: Method

```
def setdefault(self, key: builtins.str, default: typing.Optional[builtins.str] = None) -> builtins.str:
```

key

Kind: Parameter

key: builtins.str

default

Kind: Parameter

default: typing.Optional[builtins.str] = None

Default: None

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:
```

Exhaustive reference · Source

Hedge

Half-edge identifier.

Half-edge identifier.

```
class Hedge:
```

Requires features: extension-module, abi3-py310

Inspect the registered export

```
from linnet_py import Hedge
help(Hedge)
```

value

Kind: Getter

```
def value(self) -> builtins.int:
```

Numeric value of this hedge.

__new__

Kind: AssociatedFunction

```
def __new__(cls, value: builtins.int) -> Hedge:
```

Create a hedge from a zero-based index.

value

Kind: Parameter

value: builtins.int

__int__

Kind: Method

```
def __int__(self) -> builtins.int:  
Convert to int.
```

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:  
Debug-style representation.
```

[Exhaustive reference](#) · [Source](#)

HedgeNode

Hedge node with internal graph and hairs.

Hedge node with internal graph and hairs.

```
class HedgeNode:
```

Requires features: extension-module, abi3-py310

Inspect the registered export

```
from linnet_py import HedgeNode  
help(HedgeNode)
```

internal_graph

Kind: Getter

```
def internal_graph(self) -> Subgraph:  
Internal subgraph.
```

hairs

Kind: Getter

```
def hairs(self) -> Subgraph:  
Hair subgraph.
```

__new__

Kind: AssociatedFunction

```
def __new__(cls, internal_graph: Subgraph, hairs: Subgraph) -> HedgeNode:  
Create a hedge node from internal graph and hairs subgraphs.
```

internal_graph

Kind: Parameter

internal_graph: Subgraph

hairs

Kind: Parameter

hairs: Subgraph

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:  
Debug-style representation.
```

[Exhaustive reference](#) · [Source](#)

HedgePair

Hedge pairing (paired, unpaired, split).

Hedge pairing (paired, unpaired, split).

class HedgePair:

Requires features: extension-module, abi3-py310

Inspect the registered export

```
from linnet_py import HedgePair
help(HedgePair)
```

kind

Kind: Getter

```
def kind(self) -> builtins.str:
```

Kind of hedge pair: \"paired\", \"unpaired\", or \"split\".

source

Kind: Getter

```
def source(self) -> typing.Optional[Hedge]:
```

Source hedge (paired/split).

sink

Kind: Getter

```
def sink(self) -> typing.Optional[Hedge]:
```

Sink hedge (paired/split).

hedge

Kind: Getter

```
def hedge(self) -> typing.Optional[Hedge]:
```

Hedge (unpaired only).

flow

Kind: Getter

```
def flow(self) -> typing.Optional[Flow]:
```

Flow for unpaired hedge.

split

Kind: Getter

```
def split(self) -> typing.Optional[Flow]:
```

Which side is split for split edges.

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

NodeIndex

Node index identifier.

Node index identifier.

class NodeIndex:

Requires features: extension-module, abi3-py310

Inspect the registered export

```
from linnet_py import NodeIndex
help(NodeIndex)
```

value

Kind: Getter

def value(self) -> builtins.int:

Numeric value of this node index.

__new__

Kind: AssociatedFunction

def __new__(cls, value: builtins.int) -> NodeIndex:

Create a node index from a zero-based index.

value

Kind: Parameter

value: builtins.int

__int__

Kind: Method

def __int__(self) -> builtins.int:

Convert to int.

__repr__

Kind: Method

def __repr__(self) -> builtins.str:

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

NodeStatements

Dict-like proxy for node attribute statements.

Dict-like proxy for node attribute statements.

class NodeStatements:

Requires features: extension-module, abi3-py310

Inspect the registered export

```
from linnet_py import NodeStatements
help(NodeStatements)
```

__getitem__

Kind: Method

```
def __getitem__(self, key: builtins.str) -> builtins.str:
```

key

Kind: Parameter

key: builtins.str

__setitem__

Kind: Method

```
def __setitem__(self, key: builtins.str, value: builtins.str) -> None:
```

key

Kind: Parameter

key: builtins.str

value

Kind: Parameter

value: builtins.str

__delitem__

Kind: Method

```
def __delitem__(self, key: builtins.str) -> None:
```

key

Kind: Parameter

key: builtins.str

__contains__

Kind: Method

```
def __contains__(self, key: builtins.str) -> builtins.bool:
```

key

Kind: Parameter

key: builtins.str

__len__

Kind: Method

```
def __len__(self) -> builtins.int:
```

__iter__

Kind: Method

```
def __iter__(self) -> typing.Iterator[builtins.str]:
```

get

Kind: Method

```
def get(self, key: builtins.str, default: typing.Any = None) -> typing.Any:
```

key

Kind: Parameter

key: builtins.str

default

Kind: Parameter

default: typing.Any = None

Default: None

keys

Kind: Method

def keys(self) -> builtins.list[builtins.str]:

values

Kind: Method

def values(self) -> builtins.list[builtins.str]:

items

Kind: Method

def items(self) -> builtins.list[tuple[builtins.str, builtins.str]]:

clear

Kind: Method

def clear(self) -> None:

update

Kind: Method

def update(self, other: typing.Mapping[builtins.str, builtins.str]) -> None:

other

Kind: Parameter

other: typing.Mapping[builtins.str, builtins.str]

pop

Kind: Method

def pop(self, key: builtins.str, default: typing.Any = None) -> typing.Any:

key

Kind: Parameter

key: builtins.str

default

Kind: Parameter

default: typing.Any = None

Default: None

popitem

Kind: Method

def popitem(self) -> tuple[builtins.str, builtins.str]:

setdefault

Kind: Method

```
def setdefault(self, key: builtins.str, default: typing.Optional[builtins.str] = None) -> builtins.str:
```

key

Kind: Parameter

key: builtins.str

default

Kind: Parameter

default: typing.Optional[builtins.str] = None

Default: None

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:
```

Exhaustive reference · Source

Orientation

Edge orientation (Default/Reversed/Undirected).

Edge orientation (Default/Reversed/Undirected).

class Orientation:

Requires features: extension-module, abi3-py310

Inspect the registered export

```
from linnet_py import Orientation
help(Orientation)
```

default

Kind: AssociatedFunction

```
def default() -> Orientation:
```

Orientation::Default.

reversed

Kind: AssociatedFunction

```
def reversed() -> Orientation:
```

Orientation::Reversed.

undirected

Kind: AssociatedFunction

```
def undirected() -> Orientation:
```

Orientation::Undirected.

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

OrientedCut

Oriented cut represented by left/right subgraphs.

Oriented cut represented by left/right subgraphs.

```
class OrientedCut:
```

Requires features: extension-module, abi3-py310

Inspect the registered export

```
from linnet_py import OrientedCut
help(OrientedCut)
```

left

Kind: Getter

```
def left(self) -> Subgraph:
```

Left side of the cut.

right

Kind: Getter

```
def right(self) -> Subgraph:
```

Right side of the cut.

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

Subgraph

Subgraph represented as a bitset of hedges.

Subgraph represented as a bitset of hedges.

```
class Subgraph:
```

Requires features: extension-module, abi3-py310

Inspect the registered export

```
from linnet_py import Subgraph
help(Subgraph)
```

empty

Kind: AssociatedFunction

```
def empty(cls, size: builtins.int) -> Subgraph:
```

Create an empty subgraph of the given size (number of hedges).

size

Kind: Parameter

size: builtins.int

full**Kind:** AssociatedFunction

```
def full(cls, size: builtins.int) -> Subgraph:
```

Create a full subgraph including all hedges.

size**Kind:** Parameter

size: builtins.int

from_hedges**Kind:** AssociatedFunction

```
def from_hedges(cls, size: builtins.int, hedges: typing.Sequence[Hedge]) -> Subgraph:
```

Create a subgraph from a list of hedges.

size**Kind:** Parameter

size: builtins.int

hedges**Kind:** Parameter

hedges: typing.Sequence[Hedge]

to_hedges**Kind:** Method

```
def to_hedges(self) -> builtins.list[Hedge]:
```

List included hedges.

size**Kind:** Method

```
def size(self) -> builtins.int:
```

Total number of hedges in the parent graph.

n_included**Kind:** Method

```
def n_included(self) -> builtins.int:
```

Number of included hedges.

includes**Kind:** Method

```
def includes(self, hedge: typing.Any) -> builtins.bool:
```

Whether a hedge is included.

hedge**Kind:** Parameter

hedge: typing.Any

union

Kind: Method

```
def union(self, other: Subgraph) -> Subgraph:
```

Union with another subgraph.

other

Kind: Parameter

other: Subgraph

intersection

Kind: Method

```
def intersection(self, other: Subgraph) -> Subgraph:
```

Intersection with another subgraph.

other

Kind: Parameter

other: Subgraph

sym_diff

Kind: Method

```
def sym_diff(self, other: Subgraph) -> Subgraph:
```

Symmetric difference with another subgraph.

other

Kind: Parameter

other: Subgraph

subtract

Kind: Method

```
def subtract(self, other: Subgraph) -> Subgraph:
```

Subtract another subgraph.

other

Kind: Parameter

other: Subgraph

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

Exhaustive reference · Source

TraversalTree

Traversal tree produced by DFS/BFS.

Traversal tree produced by DFS/BFS.

```
class TraversalTree:
```

Requires features: extension-module, abi3-py310

Inspect the registered export

```
from linnet_py import TraversalTree
help(TraversalTree)
```

tree_subgraph

Kind: Method

```
def tree_subgraph(self) -> Subgraph:
    Subgraph corresponding to the tree edges.
```

node_order

Kind: Method

```
def node_order(self) -> builtins.list[NodeIndex]:
    Node order from traversal.
```

covers

Kind: Method

```
def covers(self, subgraph: Subgraph) -> Subgraph:
    Covers a subgraph with the traversal tree.
```

subgraph

Kind: Parameter

subgraph: Subgraph

iter_hedges

Kind: Method

```
def iter_hedges(self) -> builtins.list[tuple[Hedge, builtins.str,
typing.Optional[Hedge]]]:
    Iterate hedges as (hedge, kind, root_hedge).
```

[Exhaustive reference](#) · [Source](#)

Canonical package changelog

The following release history is imported as typed Markdown from crates/linnet/CHANGELOG.md.

Changelog

All notable changes to this project will be documented in this file.

The format is based on Keep a Changelog, and this project adheres to Semantic Versioning.

[Unreleased]

0.17.0 - 2026-01-28

Other

- force-directed layout
- make the powerset iterator properly size the subset
- upgrade to all spanning **forests** generation
- Refactor PowersetIterator to support generic ID type
- double ended iterator for subsets
- remove unused crates

0.16.1 - 2026-01-13

Other

- Quote string values in graph and parser outputs
- revert edge, node and hedge index order to be direct (not based on sort)

0.16.0 - 2026-01-08

Other

- Add methods for identifying bridges and non-bridges in subgraphs
- Make `trace_unfold` implementable outside of linnet

0.15.1 - 2026-01-05

Fixed

- fixed ordering problem in merge function and make ordering based on sorting not on direct map

Other

- update quote stripping
- Add agent context files and update project tooling
- Add graph algorithms module with topological sort and transitive ops

0.15.0 - 2025-12-08

Other

- Refactor Figment handling and improve data merging logic
- **(clinnet)** release v0.1.8

0.14.1 - 2025-12-03

Other

- Add parallel processing and directional constraints
- update test

- only swap one edge when connecting identities

0.14.0 - 2025-11-28

Other

- change output name and make it short, and add directional groups

0.13.3 - 2025-11-24

Other

- update symbolica

0.13.2 - 2025-11-12

Other

- update to symbolica 0.20
- better templates
- clinnet for nested dots to pdf
- Add Figment integration for configuration handling

0.13.1 - 2025-11-05

Other

- remove prints

0.12.0 - 2025-08-18

Other

- all tests pass
- clippy fixes
- write global data outside of graph[] scope
- Add ID=ID statements to global data when parsing
- newline after nodes
- dot serialize with name
- write graph name
- correctly parse orientations
- make iter_edges iterate with edge_id order not involution order

0.11.0 - 2025-07-29

Other

- serialize global data

0.10.0 - 2025-07-29

Other

- Add check graph function and extracting by nodes
- Add is_empty to Swap trait and update implementations

0.9.1 - 2025-07-28

Other

- impl asref for newtypes

0.9.0 - 2025-07-27

Other

- Allow specifying a subgraph in the DOT format.
- Rewrite DOT output to include node, hedge and edge indices and fix bidirectional serialization
- DotGraph has concrete data. Parser now supports global edge and vertex data. Can set edge, vertex and hedge index
- add multi-graph parsing and global graph data

0.8.0 - 2025-07-14

Other

- clippy fix
- Rename HedgeVec to EdgeVec and introduce HedgeVec
- Add H parameter to HedgeGraph in iter_edges and orientation
- add H to SignedCycle
- rename as_ref to to_ref
- Add hedge_data to hedge_graph
- add perm by key
- update symbolica
- Fix tree traversal and modify ForestNodeStore API to use Option

0.7.1 - 2025-05-24

Other

- Fix crown computation by inverting subgraph inclusion check and add inclusion for HedgePair

0.7.0 - 2025-05-23

Other

- unify iterator signature and remove redundant and misleading functions
- Add zenodo doi

0.6.3 - 2025-05-23

Other

- improve README

0.6.2 - 2025-05-22

Other

- remove useless check and set
- Correctly forget identification history and adds subgraph contraction

0.6.1 - 2025-05-22

Other

- Merge pull request #11 from alpha00p/jules_wip_3253120278989361716

0.6.0 - 2025-05-22

Fixed

- fix all tests and update symbolica

- fixes forget_identification_history, If you delete a subgraph you need

Other

- Add crown functions and implement spanning tree enumeration algo based on graph contraction
- PartialEq and Eq for hedgevec
- from_raw for hedgevec

0.5.2 - 2025-05-09

Fixed

- fix involution extraction

Other

- Add Permutation iterators, new graph extract test

0.5.1 - 2025-05-05

Other

- Clean up debug prints and add cycle test

0.5.0 - 2025-05-05

Fixed

- fixes all around

Other

- boundary->crown
- hairs->boundary

0.4.0 - 2025-04-30

Added

- feature flag bincode and serde

0.3.0 - 2025-04-30

Fixed

- fixed extraction for childvec forest, all tests pass and fix warnings

Other

- adds extract to SmartHedgeVec and involution
- maps now take a FnMut
- derive more (Decode and Encode)

0.2.3 - 2025-04-08

Other

- unpin serde

0.2.2 - 2025-04-01

Fixed

- fix test snapshots

Other

- raw image

0.2.1 - 2025-04-01

Other

- Update README.md, logo is a github asset
- put crates badge

0.2.0 - 2025-04-01

Fixed

- fix cetz orientation

Other

- Create release-plz.yml

Version information

Save these identifiers with published results and include them in issue reports so that collaborators can reproduce the same software environment.

- **Documentation channel:** latest
- **Documentation route:** /products/linnet/latest/
- **Source revision:** d909b44c9bfe20ddd13e720a7d1f18e0571254f

Package versions and enabled features

- gammaloop/gammalooprs — 0.3.4 (no optional features)
- gammaloop/gammaloop-api — 0.1.0 (features: cli)
- gammaloop/gammaloop — 0.3.4 (features: python_api, python_abi, python_stubgen)
- linnet/linnet — 0.17.0 (features: nodestore-vec, serde, bincode, drawing, symbolica)
- linnet/linnet-py — 0.1.0 (features: extension-module, abi3-py310)
- spenso/spenso — 0.6.0 (features: shadowing)
- spenso/spenso-macros — 0.3.2 (features: shadowing)
- spenso/spenso-hep-lib — 0.1.7 (no optional features)
- spenso/spynso3 — 0.1.2 (features: python_stubgen)
- idenso/idenso — 0.3.0 (features: bincode, reference-cases)
- idenso/idenso — 0.3.0 (features: python, python_stubgen)
- vakint/vakint — 0.1.2 (no optional features)
- vakint/vakint — 0.1.2 (features: symbolica_community_module, python_stubgen)